AI STRATEGY

Boehringer Ingelheim

Unlocking €1B+ in Value Through
Intelligent Automation & AI

Strategic Analysis | 2026

THE AI IMPERATIVE

Why Pharma AI Can't Wait

$2.3B

Average cost to bring a drug to market

AI cuts that through smarter R&D

90%

Clinical trial failure rate

Earlier AI prediction prevents waste

40%

Annual growth in AI drug discovery

Competitors are moving fast

AstraZeneca, Roche, Pfizer and Novartis have all established dedicated AI drug-discovery units. BI can lead, not follow.

COMPANY OVERVIEW

Boehringer Ingelheim at a Glance

Company Facts

- Founded 1885, Ingelheim, Germany
- Fully privately held (Boehringer family)
- ~€24B annual revenue (2023)
- ~53,000 employees in 130+ countries
- ~€5.7B R&D spend (≈23% of revenue)
- Therapeutic focus: Respiratory, CNS, Cardiovascular, Oncology, Animal Health
- Strong biologics manufacturing capabilities

AI Readiness Factors

✓ Private ownership — long investment horizon for AI

✓ Decades of proprietary clinical trial data

✓ Existing bioinformatics & comp-chem teams

✓ Biologics manufacturing at scale

✓ Animal Health data = unique cross-species biology

~ Data infrastructure needs unification across systems

STRATEGIC FRAMEWORK

Two Levers, One Strategy

COST REDUCTION

€300–500M annual potential

- Drug discovery & lead optimization
- Clinical trial recruitment & operations
- Manufacturing quality control
- Supply chain & demand forecasting
- Regulatory submission automation

REVENUE GROWTH

€1–2B NPV pipeline uplift

- Precision medicine & biomarker discovery
- Pipeline acceleration (2–3 years faster)
- AI-designed novel drug modalities
- Commercial excellence & sales AI
- Contract AI biomanufacturing revenue

COST REDUCTION

Where AI Cuts Costs

🧪 Drug Discovery & R&D

€80-150M / drug program

- In silico ADMET prediction cuts wet-lab iterations
- Generative AI for novel molecular scaffold design
- Target ID from multi-omics data streams
- 30-40% reduction in lead optimization time

🏥 Clinical Trial Operations

20-30% faster recruitment

- NLP to match patients from EHR & claims data
- AI protocol design to reduce amendments
- Synthetic control arms to reduce trial size
- Real-time safety signal detection at scale

🏭 Manufacturing Quality Control

€50-100M / year

- Predictive QC: catch batch failures 48hrs early
- Process analytical technology + ML integration
- Yield optimization via reinforcement learning
- Reduce OOS (out-of-spec) batch rate by 40-60%

📦 Supply Chain & Logistics

15-20% inventory savings

- AI demand forecasting across 200+ markets
- Cold chain optimization for biologics
- Supplier risk prediction & pre-emptive actions
- Reduce API write-offs by 25%+

REVENUE GROWTH

Where AI Creates Revenue

Precision Medicine

€500M+ NPV potential

- AI biomarker discovery for companion diagnostics
- Patient stratification drives better trial success
- Premium pricing for validated precision products
- Example: expand OFEV biomarker-defined populations

Pipeline Acceleration

2-3 years to market sooner

- AI compresses discovery phase by up to 50%
- Each year saved = ~€200-500M in pipeline NPV
- First-mover advantage in emerging target classes
- Screen 5-10x more compounds per dollar spent

Novel Drug Modalities

Opens new segments

- AI-designed ADCs, PROTACs, RNA therapeutics
- Generative models find chemistry humans miss
- BI biologics platform x AI = differentiated pipeline
- Access 'undruggable' targets via new modalities

Commercial Excellence

5-15% revenue uplift

- AI-driven sales territory optimization
- Dynamic pricing & market access modeling
- HCP next-best-action recommendations
- Predict adherence gaps and intervene proactively

PRIORITIZATION

Impact vs. Speed to Value

STRATEGIC BETS
(Invest for the Long Term)

- De novo AI molecular design
- Precision medicine platform
- AI-first drug discovery lab

START HERE (High Impact + Quick Win)

- Manufacturing QC AI
- Clinical trial patient matching
- ADMET prediction models
- Pharmacovigilance NLP

DEPRIORITIZE

- Generic chatbots / help desks
- Non-core HR automation

QUICK WINS (smaller impact)

- Supply chain forecasting
- Commercial analytics dashboards
- Regulatory doc drafting AI

<-- Hard to execute | SPEED TO VALUE | Easy to execute -->

YEAR 1 PRIORITIES

What to Implement First

01

⍰ AI-Augmented Lead Optimization

Highest ROI per € invested — build on existing comp-chem teams

Timeline	Investment	Expected Return
6-9 months	€15-25M	€80-120M / program

02

⍰ Clinical Trial Patient Matching

BI runs 100+ trials/year; faster recruitment = millions saved + earlier revenue

Timeline	Investment	Expected Return
4-6 months	€5-10M	20-25% recruitment savings

03

⍰ Manufacturing Predictive QC

Large biologics plant footprint already collects rich sensor data — low integration cost

Timeline	Investment	Expected Return
6-12 months	€10-20M	€50-100M / year

TRANSFORMATION ROADMAP

36-Month AI Roadmap

Months 0–12

BUILD THE FOUNDATION

- Appoint Chief AI Officer
- Launch 3 Year-1 pilots simultaneously
- Unified data lake (R&D + clinical + mfg)
- AI governance policy & ethics board
- €50M innovation fund allocated

Months 12–24

SCALE & INTEGRATE

- Roll out successful pilots enterprise-wide
- AI Center of Excellence (50+ AI staff)
- 5 additional use cases activated
- Commercial AI platform deployed
- First AI-discovered compound in preclinical

Months 24–36

TRANSFORM

- AI-native drug discovery at full scale
- 3+ AI-designed compounds in clinical trials
- Manufacturing: €100M+ savings/year
- External AI ventures & licensing revenue
- Recognised as top-5 pharma AI leader

FINANCIAL IMPACT

Estimated Value at Stake

Conservative estimates based on top-quartile pharma AI benchmarks & peer disclosures

?

€300–500M

Annual Cost Savings

Manufacturing + Clinical Ops + R&D efficiency gains

2–3 Years

Pipeline Acceleration

Earlier approvals = years of additional peak-sales exclusivity

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€1–2B

NPV Pipeline Uplift

From 2-3 programs reaching market faster + precision premium

Total 5-Year Value: €3–6B | Required Investment: €150–300M | Expected ROI: 10–20x

GOVERNANCE & RISK

Making AI Work Safely at Scale

Key Risks

Regulatory Uncertainty

FDA/EMA guidance on AI/ML in drug development still evolving

Data Quality & Silos

Fragmented legacy systems across 130+ countries

Model Explainability

Regulators need interpretable decisions for clinical use

IP & Data Privacy

GDPR compliance; proprietary data leakage risk via LLMs

Talent Competition

Competing with Big Tech for top AI/ML scientists

Mitigations

Regulatory Affairs AI Unit

Proactive FDA engagement; IPRP AI working group participation

Data Modernisation Program

CTO-sponsored 3-year roadmap to unified data lake

Explainable AI (XAI) by Default

Interpretability layer required on all clinical-facing models

Secure AI Infrastructure

On-premise LLMs for sensitive data; strict vendor AI contracts

AI Academy & Partnerships

BI AI fellowship + research ties with ETH, MIT, Cambridge

RECOMMENDATIONS

The Path Forward

1. Appoint a Chief AI Officer (CEO direct report) — signal strategic commitment and unify fragmented efforts
2. Allocate €50M innovation fund in Q1 2026 — launch the three Year-1 priorities simultaneously, not sequentially
3. Build a proprietary AI data platform — BI's private status means no external pressure to share data, a real moat
4. Partner selectively with AI-native discovery firms (Recursion, Isomorphic Labs, Insilico Medicine) to accelerate
5. Establish AI Ethics & Governance Board in Q1 — pre-empt regulatory scrutiny, earn HCP and patient trust

First action: Convene AI Strategy Working Group by April 2026 — CDO, CMO, Head of R&D, CFO